MPS Compressed Majorana Propagation

Thermal and real-time fermionic evolution with parity charges on the MPS bonds

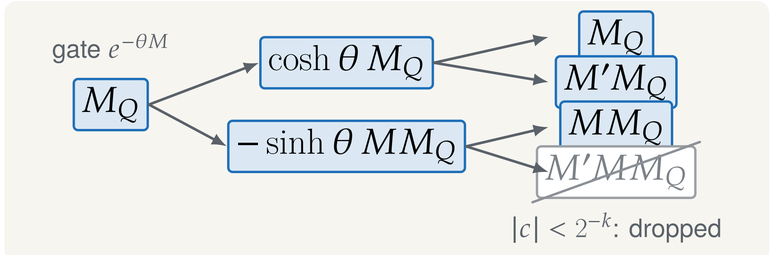
Hirmay Sandesara, Benjamin Decker, Prof. Christian Mendl

Master's thesis, Computational Science and Engineering, Technical University of Munich

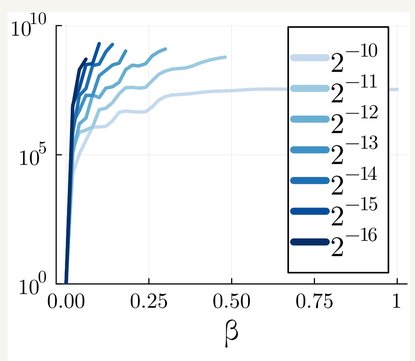
Tensor Network based approaches
to Quantum Many-Body Systems
Isaac Newton Institute, Cambridge
October 2026

1 Motivation

Heisenberg-like Majorana propagation [1, 2]: Fermionic operators expand into Majorana strings: $M_m = \mathrm{i}^{r_m}\gamma_0^{b_0}\cdots\gamma_{2N-1}^{b_{2N-1}}$ as $O = \sum_m c_m M_m$. Trotterized evolution ($O \mapsto e^{\mathrm{i}\theta M} O e^{-\mathrm{i}\theta M}$/$e^{-\theta M} O$ for real/imaginary time) branches each string into at most two (M_m and MM_m) tracked in a sparse dictionary by dropping small $|c_m|$.



Thermal string explosion: Thermal simulation evolves $I \mapsto e^{-\beta H} I$, branching simultaneously across the entire lattice. As β grows, sparsity is lost: on the 37-site Hubbard hexagon [1, App. J, Fig. 5], lowering the cutoff triggers an exponential explosion of strings, hitting **billions of strings** and a 200 GB memory limit in my replica before the energy converges.

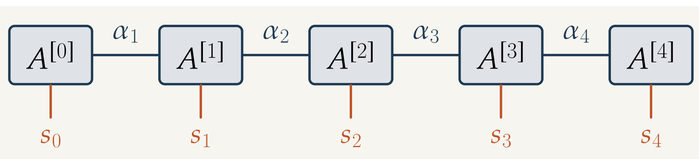


stored strings against β on 37 sites, reproduced from [1, Fig. 5]

Core question: Instead of storing independent strings and truncating $|c_m|$, can we fold the 4^N coefficients into an **operator MPS** and truncate **Schmidt values** instead?

2 The coefficients as an MPS

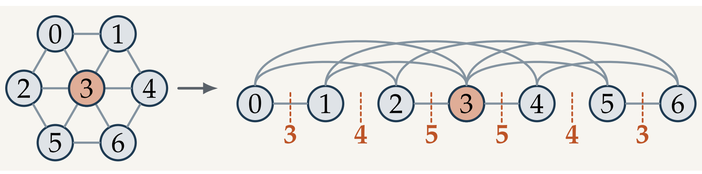
Each fermionic mode has four operator labels ($d = 4$). Their raw coefficients form an MPS:



$$a_{s_0\cdots s_{N-1}} = A^{[0]}_{s_0}\cdots A^{[N-1]}_{s_{N-1}}, \qquad s_k \in \{I,\ \gamma_{2k},\ \gamma_{2k+1},\ \gamma_{2k}\gamma_{2k+1}\}.$$

With isometric environments, the discarded singular-value norm is the operator's normalized Hilbert–Schmidt error. Operator MPS / MPDO [3, 4]; fermionic MPS [5].

3 A 2D lattice in a 1D chain



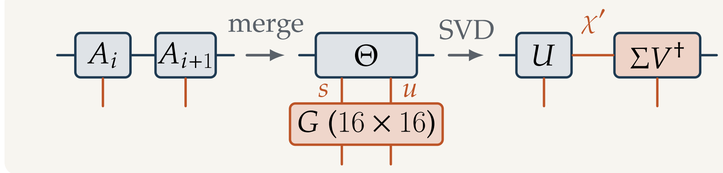
Rudolph et al.'s triangular hexagons [1], ordered row by row. Lattice edges become long-range links in the MPS. Spinful sites: two $d = 4$ tensors, or one $d = 16$ tensor. Grouping the spins removes the on-site U cut.
The widest cuts cross **5, 9 and 13 lattice edges** for 7, 19 and 37 sites. More crossing links can require larger bonds. The chain ordering matters.

Why not just cap the string length?

Length truncation assumes long strings are rare or don't feed back. At the temperatures we want, long strings are the typical ones. They're correlated, not small, so the MPS compresses their correlations instead of dropping them.

4 Short range: merge, gate, split

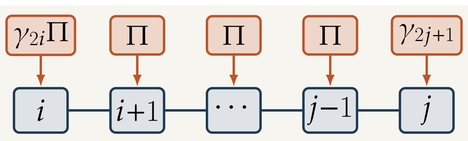
QR prepares the environments. A 16×16 superoperator then updates the two neighbouring physical labels:



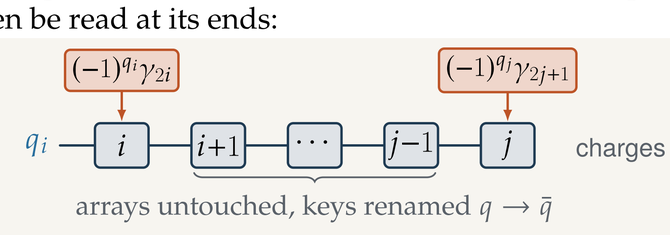
Choose the smallest rank with $\left(\sum_{a>\chi'} \sigma_a^2\right)^{1/2} \le \varepsilon$. The bond cap can require further truncation. Reuse the moving centre for the next gate; ε controls the local compression loss.

5 Long range: endpoint signs

A distant hopping carries a parity string. Here Π gives -1 to an odd local label:



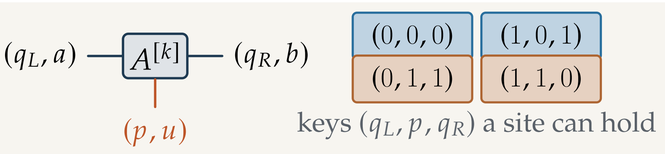
Store each virtual index as (q, a): **prefix parity** $q \in \{0, 1\}$ and a position a within that sector. The interval sign can then be read at its ends:



The intermediate charges cancel:

$$p(s_k) = q_k \oplus q_{k+1} \implies \bigoplus_{k=i}^{j-1} p(s_k) = q_i \oplus q_j.$$

Only four allowed blocks: $q_R = q_L \oplus p$, with local parity p.



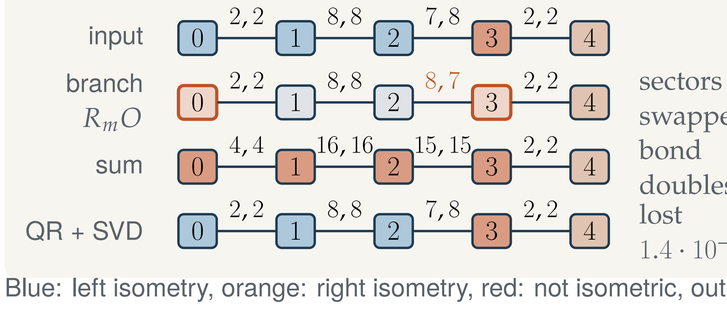
6 A long-range gate

For $M^2 = I$, write $M = R_m/\phi$ with raw string R_m and $\phi \in \{1, -\mathrm{i}\}$:

$$e^{-\theta M} O = \cosh\theta\, O - \frac{\sinh\theta}{\phi} R_m O,$$

Real time uses four branches. Add them over the affected interval, then QR and SVD each internal cut. One cutoff and one cap cover both charges.

Worked thermal step for $\gamma_0\gamma_7$; bond labels are (d_0, d_1).



Blue: left isometry, orange: right isometry, red: not isometric, outlined: multiplied. Thermal step, $\theta = -0.004$, $\varepsilon = 10^{-8}$.

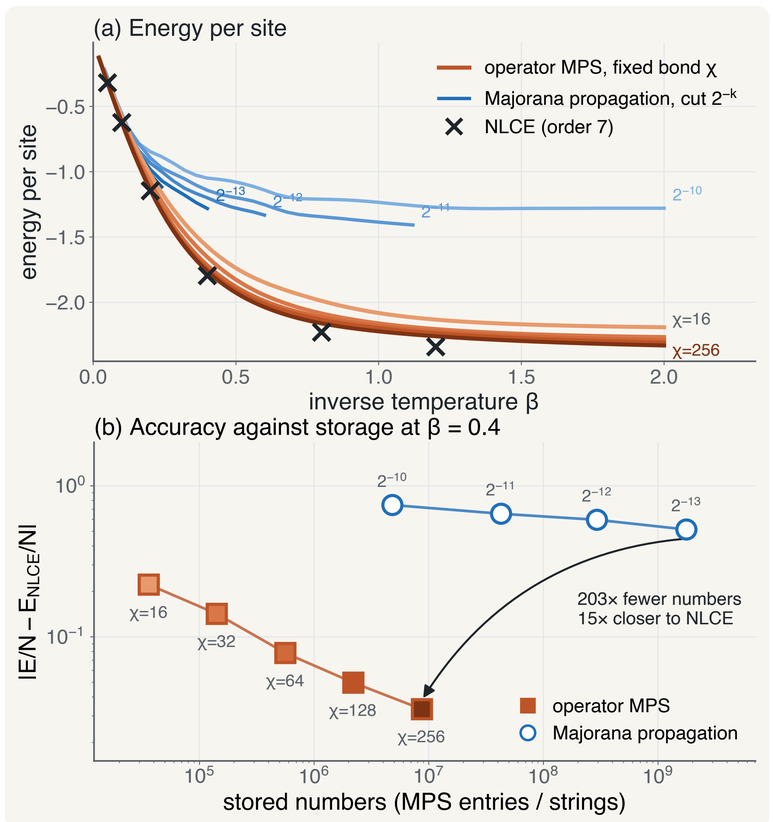
Signs are evaluated at the endpoints; compression still spans the interval. Untruncated runs agree with an independent physical-matrix Trotter reference.

Ask me about

- endpoint signs and interval compression
- $d = 4$ against $d = 16$ for spinful sites
- why real time grows the rank so fast

7 Thermal: 37-site hexagon

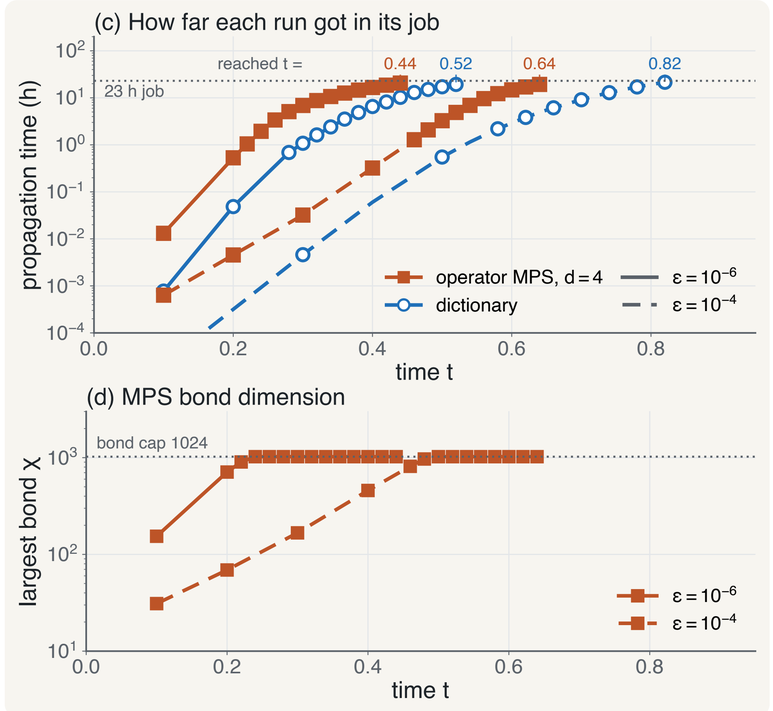
37-site spinful Hubbard hexagon, $t = 1$, $U = 8$, $\mu = U/2$: a $d = 4$ MPS against my replica of the dictionary scheme [1], with order-7 NLCE [6] as reference.



MPS rescaled by its identity coefficient after each gate; second order, $\Delta\beta = 0.02$. Replica: cuts down to 2^{-16}; first order, $\Delta\beta = 0.04$. Physical β is twice the axis in Ref. [1]. On 7 sites both splits give energy errors below 1.1×10^{-3} per site. 8–16 threads, one run per setting.

8 Real time: rapid rank growth

7-site spinful hexagon, $O(0) = 2n_{\mathrm{centre},\uparrow} - I$, $\Delta t = 0.02$, one thread.

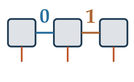


At $\varepsilon = 10^{-6}$ the MPS still stores about 4 times fewer numbers at $t = 0.4$, but once its bond is capped it falls behind. **Here runtime becomes the limit.** Both have density-row L_2 error 4.4×10^{-4} against exact evolution at $t = 0.4$, mainly from the time step. Shared nodes: reach, not speed.

Conclusion

- Many strings can fit in a moderate-rank operator MPS.
- Virtual parity charges supply the long-range signs.
- Compression across the interval is still costly; real-time rank growth is the harder limit here.

Future directions: PEPS, SWAP network, and different Hamiltonians.



No Jordan–Wigner strings were harmed in the making of this poster. They moved onto the bonds.

[1] M. S. Rudolph, A. Angrisani, A. Wright, I. Sanderski, R. Puig, Z. Holmes, *Thermal State Simulation with Pauli and Majorana Propagation*, arXiv:2602.04878 (2026).
[2] A. Miller *et al.*, *Simulation of Fermionic circuits using Majorana Propagation*, arXiv:2503.18939 (2025).
[3] F. Verstraete, J. J. Garcia-Ripoll, J. I. Cirac, *Matrix Product Density Operators: Simulation of finite-T and dissipative systems*, Phys. Rev. Lett. 93, 207204 (2004).
[4] M. Zwolak, G. Vidal, *Mixed-state dynamics in one-dimensional quantum lattice systems: a time-dependent superoperator renormalization algorithm*, Phys. Rev. Lett. 93, 207205 (2004).
[5] N. Bultinck, D. J. Williamson, J. Haegeman, F. Verstraete, *Fermionic Matrix Product States and One-Dimensional Topological Phases*, Phys. Rev. B 95, 075108 (2017).
[6] M. Rigol, T. Bryant, R. R. P. Singh, *Numerical Linked-Cluster Approach to Quantum Lattice Models*, Phys. Rev. Lett. 97, 187202 (2006).

Computations on CoolMUC-4, Leibniz Supercomputing Centre.

hirmay.github.io
Seeking a PhD position from 2027